revised law of warpage—still slightly dependent on a special class of reference frames, but less so than before.

Remaining dissatisfied, Einstein struggled all the next week with his 4 November law, found mistakes, and presented yet another proposal for the law of warpage at the Academy meeting of 11 November. But still the law relied on special frames; still it violated his principle of relativity.

Resigning himself to this violation, Einstein struggled during the next week to compute consequences of his new law that could be observed with telescopes. It predicted, he found, that starlight passing the limb of the Sun should be deflected gravitationally by an angle of 1.7 seconds of arc (a prediction that would be verified four years later by careful measurements during a solar eclipse). More important to Einstein, the new law yielded the correct perihelion shift for Mercury! He was beside himself with joy; for three days he was so excited that he couldn't work. This triumph he presented at the next meeting of the Academy on 18 November.

But his law's violation of the relativity principle still troubled him.So during the next week Einstein poured back over his calculations and found another mistake—the crucial one. At last everything fell into place. The entire mathematical formalism was now free of any dependence on special reference frames: It had the same form when expressed in each and every reference frame (see Box 2.6 below) and thus obeyed the principle of relativity. Einstein's vision of 1914 was fully vindicated! And the new formalism still gave the same predictions for the shift of Mercury's perihelion and for the gravitational deflection of light, and it incorporated his 1907 prediction of gravitational time dilation. These conclusions, and the final definitive form of his *general relativity* law of warpage, Einstein presented to the Prussian Academy on 25 November.

Three days later Einstein wrote to his friend Arnold Sommerfeld: "During the past month I had one of the most exciting and strenuous times of my life, but also one of the most successful." Then, in a January letter to Paul Ehrenfest: "Imagine my joy [that my new law of warpage obeys the principle of relativity] and at the result that the [law predicts] the correct perihelion motion of Mercury. I was beside myself with ecstasy for days." And, later, speaking of the same period: "The years of searching in the dark for a truth that one feels but cannot express, the intense desire and the alternations of confidence and misgiving until one breaks through to clarity